ReInE: Residual Inference Engine for Persona Steering in Frozen Language Models

Alethea Agung Yodha Pratama
Computer Science Department
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
alethea.pratama001@binus.ac.id

Collin Kliveson
Computer Science Department
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
collin.kliveson@binus.ac.id

Azka Dhafin Elhan
Computer Science Department
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
azka.elhan@binus.ac.id

Hidayaturrahman
Computer Science Department
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
hidayaturrahman@binus.ac.id

***Abstract*—Controlling persona, identity, and interaction style in large language models remains difficult because full fine-tuning is costly and can overwrite useful base-model behavior, while prompt-only steering is often unstable across long or adversarial interactions [1]–[5] This paper presents ReInE (*Residual Inference Engine*), a lightweight framework for steering frozen causal language models through trainable residual interventions on selected transformer layers. Rather than updating host-model weights, ReInE attaches low-rank MicroAdapters to selected blocks through forward hooks and injects additive hidden-state perturbations during both training and inference. The current implementation uses a depth-asymmetric design in which lower, middle, and upper layers are assigned different optimization priorities. Training combines assistant-token cross-entropy, KL regularization against the frozen host distribution, and a cosine-similarity anchor loss that aligns lower-layer pooled hidden states with a base-model-derived latent reference. Unlike prompt tuning and weight-space PEFT methods, the proposed implementation performs region-specific gradient routing so that different layer groups receive different objective mixtures [1], [3], [6]. Experimental results from a 157-example ablation study show that a balanced configuration with light lower-layer support, moderate middle-layer support, and dominant upper-layer control yields the strongest trade-off between identity accuracy and response consistency. The resulting adapters remain compact (1.4–2.0 MB), require fewer than 1.1 million trainable parameters, and train in under two minutes on a single NVIDIA A40 GPU. These findings suggest that residual activation-space adaptation is a practical and efficient direction for persona steering in frozen language models.**

***Index Terms*—large language model, parameter-efficient adaptation, persona steering, activation steering, frozen language model, behavioral control**

I. Introduction

Large language models (LLMs) are increasingly used in applications where response quality depends not only on factual adequacy or task completion, but also on stylistic properties such as persona, tone, self-presentation, and interaction framing [2], [7]. These behavioral properties are especially important in roleplay systems, character agents, branded assistants, and personalized conversational interfaces, where the model is expected to maintain a recognizable identity over multiple turns rather than produce isolated task answers [2], [3].

Despite this growing demand, maintaining a consistent persona across different contexts remains challenging. Prior work shows that large language models do not exhibit stable persona behavior, with consistency varying across prompts, tasks, and interaction settings, and often degrading under extended or adversarial dialogue conditions [8]. These findings suggest that persona control is not an inherent property of current models requires explicit and robust control mechanisms [9], but rather an unstable emergent behavior that requires explicit and robust control mechanisms.

Existing approaches for controlling this behavioral layer remain limited. Full fine-tuning can strongly adapt a model, but it is computationally expensive and may overwrite useful general capabilities. Parameter-efficient fine-tuning (PEFT) methods such as adapters, Prefix-Tuning, Prompt Tuning, P-Tuning v2, and LoRA reduce training cost, yet they primarily operate through weight-space or prompt-space adaptation and are often studied in the context of task transfer rather than persistent persona control [1], [6], [10]–[12]. Prompt-only control is lightweight, but it can be brittle when prompts are paraphrased, extended through conversation history, or challenged by adversarial instructions [2], [3].

A separate line of work shows that model behavior can be redirected by intervening on internal activations instead of modifying the full host model. Plug-and-Play Language Models, Activation Addition, and Inference-Time Interven-

tion all demonstrate that behavior can be shifted through lightweight guidance or hidden-state edits, while more recent activation-engineering approaches further refine this paradigm through learned and contrastive steering mechanisms [13], [14]. Model-editing methods such as ROME and MEMIT show that targeted interventions inside transformer representations can affect specific associations without retraining the full network [2], [3], [15]–[17]. These findings motivate the idea that persona and identity control may also be implemented effectively in activation space.

Prior work further shows that transformer layers encode qualitatively different representations across depth, with earlier layers capturing broader semantic structure and later layers contributing more directly to token-level realization [18].

This paper introduces **ReInE** (*Residual Inference Engine*), a frozen-host framework for persona steering through trainable residual intervention in hidden-state space. ReInE does not update the base model weights. Instead, it attaches low-rank MicroAdapters to selected transformer layers through forward hooks and adds learned perturbations to the intermediate representations [1], [6]. The current implementation is explicitly *depth-asymmetric*: lower, middle, and upper layers are not treated uniformly, but are instead assigned different optimization roles through separate parameter groups and different objective weights.

The central hypothesis of ReInE is that persona steering is distributed unevenly across depth. In the current design, lower layers are encouraged to align with a latent anchor derived from the frozen host model, middle layers receive moderated cross-entropy and KL supervision, and upper layers receive the strongest output-level supervision. This design is motivated by prior evidence that hidden-state interventions at different depths can produce different behavioral effects, and by the practical observation that late layers are often closest to token-level realization while earlier layers encode broader representational structure [3], [15], [16].

To study this hypothesis, the present work focuses on controlled architectural ablation rather than large-scale benchmark comparison. Three settings are compared: *Upper Only*, *Lower Focused*, and *Full (Balanced)*. The results show that dominant upper-layer control is important for explicit identity rendering, heavy reliance on lower layers alone is less stable under multi-turn history, and the strongest overall trade-off is obtained by combining light lower-layer support, moderate middle-layer support, and stronger upper-layer intervention.

The contributions of this paper are as follows:

- It introduces ReInE, a frozen-host residual adapter framework for persona steering through activation-space intervention.
- It formalizes a depth-asymmetric training design in which lower, middle, and upper transformer regions receive different objective mixtures.
- It presents an anchor-based latent supervision mechanism that aligns lower-layer pooled hidden states with a base-model-derived reference representation.
- It identifies a compact operating regime in which dominant upper-layer control, with limited lower- and middle-layer support, yields the strongest identity accuracy and consistency in the reported ablations.

This work focuses on frozen-host persona steering rather than general factual rewriting or broad model editing. The current experiments are conducted on a reasoning-capable host model and use a compact dialogue dataset, so broader generalization across model families, training scales, and alternative anchor objectives remains future work [3], [15], [17].

II. Literature Review

Research relevant to ReInE spans parameter-efficient adaptation, activation-space steering, model editing, and controlled generation. Together, these areas suggest that useful behavioral control can often be achieved without updating all parameters of a large language model [1], [2], [6].

A. Parameter-Efficient Adaptation

Parameter-efficient adaptation methods show that a frozen backbone can be specialized through relatively small trainable modules. Classical adapters insert new bottleneck layers into a pretrained network, while AdapterFusion extends this idea toward composing multiple task adapters without destructive retraining [6], [19]. These methods established that downstream adaptation does not necessarily require updating the full host model.

Prompt-side PEFT methods further reduced the trainable footprint by learning continuous prefixes or soft prompts instead of modifying all weights. Prefix-Tuning, Prompt Tuning, and P-Tuning v2 demonstrated that lightweight learned prompt representations can become competitive with larger-scale fine-tuning under suitable conditions [10]–[12]. However, these methods still express control through prompt parameterization rather than direct intervention on hidden representations [10]–[12], [20].

LoRA, BitFit, Compacter, and IA^3 remain influential PEFT baselines [1], [21]–[24], because they achieve strong adaptation with much smaller trainable parameter counts than full fine-tuning [1], [21]–[23]. Nevertheless, these methods are still fundamentally weight-space adaptation methods. ReInE differs in operating through residual hidden-state intervention attached by hooks, while keeping the host model frozen and treating different depth regions asymmetrically.

These limitations suggest that weight-space and prompt-space adaptation alone may be insufficient for enforcing stable and persistent behavioral control across diverse interaction settings.

B. Activation Steering and Representation-Level Control

Activation-space methods provide a closer conceptual foundation for ReInE. Plug-and-Play Language Models showed that generation can be guided toward desired attributes without retraining the host model [2], [25], establishing an important precedent for lightweight control at inference time [2]. This line of work suggests that useful behavioral modification can

emerge from intervention around the generative process rather than full model replacement.

Activation Addition showed that fixed steering vectors added to hidden states can redirect topic, sentiment, and other output properties without optimization, while mean-centering refinements improved the stability of such steering vectors [3], [26]. Dynamic activation composition further extended this paradigm by combining multiple steering signals within the same generation process [27]. These works support the general idea that hidden-state manipulation is a viable control surface [3], [26], [28] for language models.

Inference-Time Intervention demonstrated that targeted hidden-state edits can improve truthfulness in a frozen model, showing that representation-level control can affect output behavior in meaningful and measurable ways [15]. ReInE differs from such methods in learning reusable trainable modules rather than manually derived or externally supplied steering vectors, but both approaches rely on the premise that internal activations can be a more direct control target than full weight updates.

Model-editing methods such as ROME and MEMIT provide additional evidence that transformer internals contain localized and manipulable behavioral or factual structure [16], [17]. Although these methods focus on editing factual associations rather than persistent persona steering, they reinforce the broader argument that selective intervention inside the network can alter specific aspects of model behavior without retraining the entire model [16], [17], [29].

Recent work extends activation steering beyond fixed-vector intervention. Activation engineering approaches demonstrate that behavior can be controlled through directions in representation space without retraining the base model [13]. Contrastive activation steering further improves precision by constructing steering directions from paired examples, enabling more targeted behavioral modification [14]. While these methods provide strong control over style and attributes, maintaining consistent persona across multi-turn dialogue remains challenging, as models may still exhibit drift or contradictions under extended interaction [8].

C. Positioning of ReInE

ReInE sits between PEFT and activation steering. Like PEFT methods, it keeps the host model frozen and learns only a small number of additional parameters. Like activation-steering methods, it applies control directly in hidden-state space instead of relying solely on prompts or broad weight updates [1], [3], [6]. Its distinguishing feature in the current implementation is not only the use of trainable residual modules, but also the assignment of different objective mixtures to different layer groups through explicit parameter grouping and gradient routing.

Unlike prompt-only approaches, ReInE is designed to produce a persistent behavioral overlay that remains attached to the model regardless of how the user phrases the prompt [2]. Unlike LoRA-style methods, its intervention occurs at the level of intermediate activations rather than by rewriting existing projection weights [1]. Unlike fixed-vector steering, it learns residual modules that can be saved, reloaded, and reused as standalone adapters [3]. This positioning makes ReInE a lightweight but explicitly trainable form of activation-space persona control.

Reinforcement Learning from Human Feedback (RLHF) has emerged as a dominant paradigm for aligning large language models with human preferences. This approach improves helpfulness and safety by optimizing model outputs against learned reward signals derived from human feedback [30]. However, RLHF introduces challenges such as reward misgeneralization, bias propagation, and vulnerability to adversarial prompting, and may still produce inconsistent behavior under distribution shift [31]. These limitations suggest that while RLHF improves alignment at a global level, it does not fully address fine-grained and persistent persona control, motivating approaches that operate directly on internal representations.

III. Methodology

A. Overview

We propose **ReInE** (*Residual Inference Engine*), a parameter-efficient framework for persona steering in frozen causal language models through depth-asymmetric residual intervention. Let M_{host} denote a pretrained causal language model with frozen parameters Θ_{host}. ReInE introduces trainable parameters Θ_{ReInE} while keeping the host unchanged:

$$\nabla\Theta_{\text{host}} = 0, \qquad \nabla\Theta_{\text{ReInE}} \neq 0.$$

The adapted model intervenes on selected hidden states rather than rewriting host weights directly:

$$M_{\text{ReInE}}(x) = M_{\text{host}}(x; \Theta_{\text{host}}) + \Delta(x; \Theta_{\text{ReInE}}),$$

where Δ denotes the cumulative residual perturbations introduced by the trainable adapters.

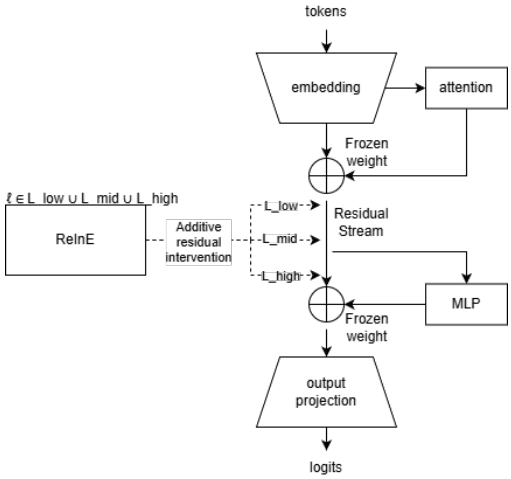

Fig. 1. Overview of ReInE residual intervention in a frozen transformer.

B. Layer-wise Residual Intervention

Let the transformer contain L layers. ReInE selects three subsets of layers,

$$\mathcal{L}_{\text{low}}, \qquad \mathcal{L}_{\text{mid}}, \qquad \mathcal{L}_{\text{high}} \subset \{0, \ldots, L-1\},$$

corresponding to lower, middle, and upper depth regions. In the reported configuration, these are implemented as lower layers $\{3, 4\}$, middle layers $\{16, 17\}$, and upper layers $\{27, \ldots, 35\}$.

Each selected layer ℓ is augmented with a low-rank MicroAdapter attached through a forward hook. Given a hidden state $h^{(\ell)} \in \mathbb{R}^{T \times d}$, the hooked representation is updated as

$$\tilde{h}^{(\ell)} = h^{(\ell)} + s^{(\ell)} f_{\theta^{(\ell)}}(h^{(\ell)}),$$

where $s^{(\ell)}$ is a learnable per-layer scale parameter. The adapter function is

$$f_{\theta^{(\ell)}}(h) = \gamma^{(\ell)} W_{\text{up}}^{(\ell)} \Big(W_{\text{down}}^{(\ell)}(\text{Dropout}(h)) \Big),$$

where $W_{\text{down}}^{(\ell)} \in \mathbb{R}^{r \times d}$, $W_{\text{up}}^{(\ell)} \in \mathbb{R}^{d \times r}$, $r \ll d$, and $\gamma^{(\ell)}$ is the adapter-internal scaling factor determined by the adapter rank and scaling hyperparameter. The up-projection is initialized to zero so that the initial model remains close to the frozen host.

C. Training Data and Token Supervision

Training data is stored in JSONL format as multi-turn dialogue. For each example, all turns except the final assistant turn are treated as context c, and the final assistant message is treated as target response y. The resulting token sequence is

$$x = [c; y; \texttt{eos}],$$

with label mask

$$\ell = [-100, \ldots, -100, y, \texttt{eos}],$$

so that cross-entropy is computed only on assistant-side tokens.

The current implementation optionally supports explicit handling of `<think>` spans, but the reported experiments disable direct supervision over those spans. In the reported configuration, `include_think = false`, so internal reasoning content is excluded from token-level loss and persona behavior is learned from the final assistant response rather than from explicit chain-of-thought imitation.

D. Anchor-Based Latent Supervision

To stabilize lower-layer behavior, ReInE introduces an auxiliary anchor objective in representation space. An *anchor object* a is first constructed from configurable text input. The implementation supports several modes, including instruction text, compiled history from the dataset, and external chat-history text. In the experiments reported here, the anchor is built from instruction text.

The frozen host model processes the anchor object, and the hidden states from the selected lower layers are captured:

$$h_{\text{anchor}}^{(\ell)} = M_{\text{host}}^{(\ell)}(a), \qquad \ell \in \mathcal{L}_{\text{low}}.$$

For each lower layer, a pooled anchor vector is obtained by masked mean pooling over the sequence:

$$v^{(\ell)} = \text{meanpool}\Big(h_{\text{anchor}}^{(\ell)} \Big).$$

During training, the adapted model also produces lower-layer hidden states for each batch. These are pooled in the same way and encouraged to align with the corresponding anchor vectors using cosine similarity:

$$\mathcal{L}_{\text{anchor}} = \frac{1}{|\mathcal{L}_{\text{low}}|} \sum_{\ell \in \mathcal{L}_{\text{low}}} \Big(1 - \cos\Big(\bar{h}^{(\ell)}, v^{(\ell)} \Big) \Big),$$

where $\bar{h}^{(\ell)}$ denotes the pooled hidden state for the current batch. Intuitively, this objective pulls lower-layer representations toward a base-model-derived latent reference rather than supervising exact output tokens.

E. Output-Level Losses

The primary output-level loss is assistant-token cross-entropy:

$$\mathcal{L}_{\text{CE}} = -\sum_{t \in \mathcal{A}} \log p_{\text{adapt}}(y_t \mid x_{<t}),$$

where $\mathcal{A}$ is the set of supervised assistant-token positions. The implementation uses weighted token-level cross-entropy, and can assign special weight to selected structural tokens such as `</think>`; however, this mechanism does not supervise hidden reasoning content directly when `include_think = false`.

To keep the adapted model close to the frozen host distribution, ReInE also applies KL regularization between the adapted and base logits:

$$\mathcal{L}_{\text{KL}} = D_{\text{KL}}\big(p_{\text{adapt}} \,\|\, p_{\text{base}} \big).$$

This regularization limits unstable drift and helps preserve fluency while still allowing controlled behavioral deviation.

F. Depth-Asymmetric Gradient Routing

The current ReInE implementation does not optimize all trainable parameters with a single uniform loss weight. Instead, trainable parameters are divided into three groups associated with lower, middle, and upper layers, and separate weighted objectives are assigned to each group:

$$\mathcal{L}_{\text{upper}} = w_{\text{CE}}^{u} \mathcal{L}_{\text{CE}} + w_{\text{KL}}^{u} \mathcal{L}_{\text{KL}},$$

$$\mathcal{L}_{\text{mid}} = w_{\text{CE}}^{m} \mathcal{L}_{\text{CE}} + w_{\text{KL}}^{m} \mathcal{L}_{\text{KL}},$$

$$\mathcal{L}_{\text{low}} = w_{\text{CE}}^{l} \mathcal{L}_{\text{CE}} + w_{\text{KL}}^{l} \mathcal{L}_{\text{KL}} + w_{\text{anchor}}^{l} \mathcal{L}_{\text{anchor}}.$$

Gradients from these group-specific objectives are computed separately and then assigned to the corresponding parameter groups before the optimizer step. In the reported balanced configuration, upper layers receive the strongest cross-entropy supervision, middle layers receive reduced cross-entropy plus KL regularization, and lower layers receive dominant anchor supervision with only light output-level pressure. This design implements the depth-asymmetric hypothesis directly at the optimization level rather than only at the architectural level.

G. Implementation Configuration

All experiments in this paper use `unsloth/Qwen3-4B-Thinking-2507` as the frozen host model. Adapters use rank $r = 16$, scaling hyperparameter $\alpha = 16$, and dropout 0.1. The reported balanced configuration initializes weaker layer scales in lower layers, moderate scales in middle layers, and stronger scales in upper layers. Training uses AdamW, cosine learning-rate scheduling, gradient clipping with maximum norm 1.0, batch size 2, and 5 epochs. Only adapter parameters and layer-scale parameters are trainable.

IV. Results

A. Evaluation Setup and Metrics

All experiments use the same frozen host model, `unsloth/Qwen3-4B-Thinking-2507`, and the same 15-prompt evaluation set spanning direct identity queries, paraphrased identity questions, role-shift instructions, and adversarial identity-override prompts. Each model is evaluated in two modes: *stateless*, where each prompt is asked independently, and *history-carried*, where prior dialogue context is retained. This distinction separates one-shot identity rendering from multi-turn persistence.

Responses are manually scored using a three-level scheme: 2 for correct, 1 for partially correct, and 0 for incorrect. Identity Accuracy is defined as

$$\text{Identity Accuracy} = \frac{\sum_i s_i}{2N}, \tag{1}$$

where $N = 15$ and $s_i \in \{0, 1, 2\}$. In addition to numeric scoring, qualitative consistency is assessed based on stability of self-identity, full-name expansion, creator attribution, and general response framing across paraphrases, adversarial prompts, and continued history.

Two training regimes are reported. The first is a compact *ablation stress test* using the same 157-example dialogue dataset across all ReInE ablation settings to isolate the contribution of lower, middle, and upper layer groups under constrained supervision. The second is a larger *comparison dataset* used for the ReInE–QLoRA comparison. This dataset contains approximately 600 examples organized into six identity-related topic groups, with at least 50 samples per topic, to provide broader semantic coverage of the target identity.

B. Ablation Results on the Compact Stress-Test Dataset

We first evaluate ReInE on the compact 157-example stress-test dataset. Three settings are compared: *Upper Only*, *Lower Focused*, and *Full (Balanced)*. This experiment is intended to isolate the functional contribution of different depth regions rather than maximize absolute performance.

Table I summarizes identity accuracy across the three settings.

TABLE I
PROMPT-LEVEL IDENTITY ACCURACY ACROSS ABLATION SETTINGS

Configuration	ZS Score	ZS Acc.	Hist. Score	Hist. Acc.
Upper Only	17/30	0.567	15/30	0.500
Lower Focused	20/30	0.667	12/30	0.400
Full (Balanced)	24/30	0.800	29/30	0.967

The balanced configuration performs best overall, especially in the history-carried setting, indicating that combined lower-, middle-, and upper-layer intervention yields the strongest persistence under continued dialogue. The lower-focused setting remains effective in stateless prompting, suggesting that lower-layer intervention captures broad semantic tendencies, but it degrades substantially under carried history. The upper-only setting is comparatively stronger for direct identity rendering but remains less stable than the balanced design. Overall, the results support a progressive depth view in which lower layers contribute early semantic shaping, middle layers refine that representation, and upper layers are most effective when the information is already close to output realization.

C. Resource Footprint of the Ablation Adapters

Table II reports the resource footprint of the three ablation adapters.

TABLE II
RESOURCE FOOTPRINT OF ABLATION ADAPTERS

Configuration	Size	Params	Time (min)	Peak VRAM (GB)
Upper Only	1.4 MB	737,298	0.94	9.98
Lower Focused	1.4 MB	737,298	1.16	10.42
Full (Balanced)	2.0 MB	1,064,986	1.52	10.36

All ablation checkpoints remain compact. Even the balanced configuration uses only about 1.06 million trainable parameters, produces a 2.0 MB checkpoint, and trains in about 1.5 minutes on a single NVIDIA A40 GPU. This indicates that activation-space persona steering can be achieved with a very small trainable footprint relative to the host model.

D. Zero-Shot Comparison with QLoRA on the Larger Six-Topic Dataset

We next compare ReInE against a QLoRA baseline using a larger identity-shift dataset of approximately 600 examples. Unlike the compact ablation setting, this dataset is organized into six topic groups: self identity, full name, creator relation, creator identity, host-model rejection, and casual or contextual dialogue behavior. Each topic contributes at least 50 samples, providing broader semantic coverage of the target identity.

Both methods are evaluated using the same 15-prompt stateless identity test and the same three-level scoring scheme. Because the goal of this comparison is zero-shot identity binding under matched supervision, only the stateless setting is reported here.

Table III summarizes the results.

TABLE III
ZERO-SHOT IDENTITY ACCURACY: REINE VS. QLORA ON THE LARGER SIX-TOPIC DATASET

Configuration	Raw Score	Raw Acc.
QLoRA (latest)	8/30	0.267
ReInE (latest)	26/30	0.867

The larger-dataset comparison shows a clear qualitative difference between the two methods. QLoRA learns parts of the target creator relation, but this adaptation remains fragmented: it still frequently falls back to the host-model identity on self-name and full-name prompts. ReInE, by contrast, maintains a much more coherent identity shift across related slots. This suggests that ReInE benefits less from repeated surface strings and more from semantically distributed supervision across multiple identity-related facets. The comparison also indicates that dataset structure matters alongside regularization: earlier trials with stronger KL weakened identity expression, but the newer six-topic dataset suggests that broader semantic coverage is also necessary for stable identity binding.

To contextualize these results, Table IV reports the resource footprint of the compared models.

TABLE IV
RESOURCE FOOTPRINT: REINE VS. QLORA ON THE LARGER SIX-TOPIC DATASET

Configuration	Size	Params	Time (min)	VRAM (GB)
QLoRA	126.1 MB	33,030,144	8.27	4.41
ReInE	4.1 MB	1,064,986	6.20	10.29

ReInE remains substantially smaller in trainable footprint, checkpoint size, and training time, while QLoRA benefits from lower peak VRAM due to 4-bit quantized loading of the host model. Accordingly, the VRAM comparison should be interpreted in light of the different base-model loading regimes rather than as a pure adapter-efficiency comparison. Overall, the results suggest that ReInE is better suited than standard QLoRA for coherent identity-shift adaptation under compact but semantically structured supervision.

E. Preliminary Knowledge-Substitution Observation

An additional qualitative observation emerged during testing. ReInE preserved dense self-lore and core identity information reasonably well, but knowledge substitution became brittle when two entities occupied nearby semantic space without equally rich relational supervision.

In the observed case, dense ReInE-specific supervision was combined with a smaller amount of factual material about Pavolia ReInE. Because the two identities share partial semantic overlap at the name level, the adapted model sometimes maintained explicit identity denial while still leaking nearby attributes under underconstrained prompts. This suggests that the current cross-entropy-plus-anchor training recipe can support partial response-level knowledge injection, but does not guarantee clean separation between semantically neighboring identities under sparse supervision. More broadly, the result reinforces the earlier finding that ReInE benefits from diverse relational coverage rather than isolated statements alone.

ACKNOWLEDGMENT

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks ...". Instead, try "R. B. G. thanks...". Put sponsor acknowledgments in the unnumbered footnote on the first page.

REFERENCES

[1] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "Lora: Low-rank adaptation of large language models," *arXiv preprint arXiv:2106.09685*, 2021.

[2] S. Dathathri, A. Madotto, J. Lan, J. Hung, E. Frank, P. Molino, J. Yosinski, and R. Liu, "Plug and play language models: A simple approach to controlled text generation," in *International Conference on Learning Representations (ICLR)*, 2020.

[3] A. M. Turner, L. Thiergart, D. Udell, G. Leech, U. Mini, and M. MacDiarmid, "Activation addition: Steering language models without optimization," *arXiv preprint arXiv:2308.10248*, 2023.

[4] E. Perez and et al., "Red teaming language models with language models," *arXiv preprint arXiv:2202.03286*, 2022.

[5] D. Ganguli and et al., "Predictability and surprise in large generative models," *arXiv preprint arXiv:2202.07785*, 2022.

[6] N. Houlsby, A. Giurgiu, S. Jastrzebski, B. Morrone, Q. de Laroussilhe, A. Gesmundo, M. Attariyan, and S. Gelly, "Parameter-efficient transfer learning for nlp," in *Proceedings of the 36th International Conference on Machine Learning (ICML)*, 2019.

[7] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. Le, and D. Zhou, "Chain-of-thought prompting elicits reasoning in large language models," *arXiv preprint arXiv:2201.11903*, 2022.

[8] M. Reusens, B. Baesens, and D. Jurgens, "Are economists always more introverted? analyzing consistency in persona-assigned llms," *arXiv preprint arXiv:2506.02659*, 2025.

[9] P. Liang and et al., "Holistic evaluation of language models," *arXiv preprint arXiv:2211.09110*, 2023.

[10] X. L. Li and P. Liang, "Prefix-tuning: Optimizing continuous prompts for generation," *arXiv preprint arXiv:2101.00190*, 2021.

[11] B. Lester, R. Al-Rfou, and N. Constant, "The power of scale for parameter-efficient prompt tuning," *arXiv preprint arXiv:2104.08691*, 2021.

[12] X. Liu, K. Ji, Y. Fu, Z. Du, Z. Yang, and J. Tang, "P-tuning v2: Prompt tuning can be comparable to fine-tuning universally across scales and tasks," *arXiv preprint arXiv:2110.07602*, 2021.

[13] A. M. Turner, L. Thiergart, G. Leech, D. Udell, J. J. Vazquez, U. Mini, and M. MacDiarmid, "Steering language models with activation engineering," *arXiv preprint arXiv:2308.10248*, 2024.

[14] N. Panickssery, N. Gabrieli, J. Schulz, M. Tong, E. Hubinger, and A. M. Turner, "Steering llama 2 via contrastive activation addition," *arXiv preprint arXiv:2312.06681*, 2024.

[15] K. Li, O. Patel, F. Viégas, H. Pfister, and M. Wattenberg, "Inference-time intervention: Eliciting truthful answers from a language model," *arXiv preprint arXiv:2306.03341*, 2023.

[16] K. Meng, D. Bau, A. Andonian, and Y. Belinkov, "Locating and editing factual associations in gpt," *arXiv preprint arXiv:2202.05262*, 2022.

[17] K. Meng, A. S. Sharma, A. Andonian, Y. Belinkov, and D. Bau, "Mass-editing memory in a transformer," *arXiv preprint arXiv:2210.07229*, 2022.

[18] O. Skean, M. R. Arefin, D. Zhao, N. Patel, J. Naghiyev, Y. LeCun, and R. Shwartz-Ziv, "Layer by layer: Uncovering hidden representations in language models," *arXiv preprint arXiv:2502.02013*, 2025.

[19] J. Pfeiffer, A. Kamath, A. Rücklé, K. Cho, and I. Gurevych, "Adapterfusion: Non-destructive task composition for transfer learning," in *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 2021.

[20] W. L. Hamilton and et al., "Promptsource: An integrated development environment and repository for natural language prompts," *arXiv preprint arXiv:2202.01279*, 2022.

[21] E. B. Zaken, S. Ravfogel, and Y. Goldberg, "Bitfit: Simple parameter-efficient fine-tuning for transformer-based masked language-models," *arXiv preprint arXiv:2106.10199*, 2021.

[22] R. K. Mahabadi, J. Henderson, and S. Ruder, "Compacter: Efficient low-rank hypercomplex adapter layers," *arXiv preprint arXiv:2106.04647*, 2021.

[23] H. Liu, D. Tam, M. Muqeeth, J. Mohta, T. Huang, M. Bansal, and C. Raffel, "Few-shot parameter-efficient fine-tuning is better and cheaper than in-context learning," *arXiv preprint arXiv:2205.05638*, 2022.

[24] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "Qlora: Efficient finetuning of quantized llms," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[25] B. Krause, A. Gotmare, B. McCann, N. S. Keskar, S. Joty, R. Socher, and N. F. Rajani, "Gedi: Generative discriminator guided sequence generation," in *International Conference on Learning Representations (ICLR)*, 2021.

[26] O. Jorgensen, D. Cope, N. Schoots, and M. Shanahan, "Improving activation steering in language models with mean-centring," *arXiv preprint arXiv:2312.03813*, 2023.

[27] D. Scalena, G. Sarti, and M. Nissim, "Multi-property steering of large language models with dynamic activation composition," *arXiv preprint arXiv:2406.17563*, 2024.

[28] N. Subramani, S. Suresh, and M. Peters, "Extracting latent steering vectors from pretrained language models," in *Proceedings of the Association for Computational Linguistics (ACL)*, 2022.

[29] E. Mitchell, C. Lin, A. Bosselut, C. D. Manning, and C. Finn, "Fast model editing at scale," in *International Conference on Learning Representations (ICLR)*, 2022.

[30] Y. Bai, A. Jones, K. Ndousse *et al.*, "Training a helpful and harmless assistant with reinforcement learning from human feedback," *arXiv preprint arXiv:2204.05862*, 2022.

[31] S. Casper, X. Davies *et al.*, "Open problems and fundamental limitations of reinforcement learning from human feedback," *arXiv preprint arXiv:2307.15217*, 2023.